

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Modular Unit 2

Adaptive Classified

OL-T29 · Sequences

Questions

Student book · questions with writing space

■ 19 pages

FAMILY MAP

Sequences

F01

Recover arithmetic-series parameters

10 Q | 50 marks

F02

Use an arithmetic nth term

7 Q | 21 marks

24 An arithmetic series has 30 terms.

The first term is a

The common difference is d

The 20th term is 123

The sum of the 30 terms is 2880

Work out the value of a and the value of d

Show clear algebraic working.

$a = \dots\dots\dots$

$d = \dots\dots\dots$

(Total for Question 24 is 5 marks)

Question 01

4MA1-1H Q1

MODULAR UNIT 2

3 marks

- 1 Here are the first four terms of an arithmetic sequence.

1 4 7 10

- (a) Find an expression, in terms of n , for the n th term of this sequence.

.....
(2)

The n th term of a different arithmetic sequence is $5n + 17$

- (b) Find the 12th term of this sequence.

.....
(1)

Question 11

4MA1-1H Q24

MODULAR UNIT 2

6 marks

24 A polygon has n sides, where $n > 5$

The interior angles of the polygon form an arithmetic sequence.

The smallest angle of the polygon is 84°

The common difference of the sequence is 4°

Work out the sum of the interior angles of the polygon.

Show clear algebraic working.

○

Question 15

4MA1-1H Q19 demand

MODULAR UNIT 2

6 marks

19 Here are the first 4 terms in an arithmetic sequence.

3 7 11 15

The last term of the sequence is x

The sum of the terms of the sequence is 7260

Find the value of x

Show clear algebraic working.

$x =$

4MA1-1H Q20

6 marks

- Show your working clearly.

$X =$

4MA1-1H Q19 M01

5 marks

- Work out the sum of the first 45 terms of the arithmetic series.
Show clear algebraic working.

.....

Question 01

4MA1-1H Q1 Q1(a)

MODULAR UNIT 2

2 marks

- 1 Here are the first five terms of an arithmetic sequence.

1 5 9 13 17

- (a) Find an expression, in terms of n , for the n th term of this sequence.

(2)

Question 02

4MA1-1H Q1 Q1(b)

MODULAR UNIT 2

1 marks

The n th term of another arithmetic sequence is $3n + 5$

(b) Find an expression, in terms of m , for the $(2m)$ th term of this sequence.

.....
(1)

Question 38

4MA1-2H Q26

MODULAR UNIT 2

5 marks

- 26 An arithmetic series has first term a and common difference d , where d is a prime number.

The sum of the first n terms of the series is S_n and

$$S_m = 39$$

$$S_{2m} = 320$$

Find the value of d and the value of m
Show clear algebraic working.

$$d = \dots\dots\dots$$

$$m = \dots\dots\dots$$

He is going to continue to save, up to and including the year 2070, by increasing the amount he saves each year by \$ k

Work out the value of k .

 $k =$

The sum of the first $2n$ terms of the series is four times the sum of the first n terms of the series.

$a =$

85 79 73 67

- Find the value of N .
Show clear algebraic working.

$$N = \dots\dots\dots$$

$$k \quad \frac{3k}{4} \quad \frac{k}{2} \quad \frac{k}{4}$$

calculate the sum of the first 30 terms of the series.

8 15 22 29 36

Work out the sum of all the terms from the 50th term to the 100th term inclusive.

- The common difference of the series is 3

The sum of the first n terms of the series can be written as $p(qt - 1)^r$ where p , q and r are integers.

Find the value of p , the value of q and the value of r
Show clear algebraic working.

(Total for Question 22 is 4 marks)

